Natural Language Processing (NLP) of Nepali Audio files and Text Analytics

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Nepal

Outline

- Need of processing Nepali audio files
- Past experience of processing Nepali audio files
- Present scenario of processing Nepali audio files
- Use of processed Nepali audio files
- Content analysis: Quantitative and Qualitative

Research paradigm

- Positivist = Quantitative – One truth (MSc in Statistics, Nepal)
- Constructivist/Interpretivist = Qualitative – multiple truths (Mphil in Demography, Mexico)
- Pragmatist = Quantiative + Qualitative (Mixed Methods) – Multiple truths (Health Professions Education, India/USA)
- Nepali audio files are **needed** for Qualitative and Mixed Methods research/projects in Public Health Sciences and Health Professions Education

Past experience

- Public health sciences and health professions education often relies on collecting rich data through structured, face-to-face interviews in “Nepali” with interview guides.

- Nepali audio file transcription required a rigorous transcription process in Nepali before advanced analysis can commence.

- It was a time consuming “manual” job. The researcher needed more time to prepare the “transcript aka text” for further steps.

Present scenario 1

- The automatic speech recognition (ASR) became available for key languages like English, Spanish etc. with good accuracy
- The OpenAI Whisper models provided the best ASR experience for these languages
- ASR of Nepali lauguage was not so accurate with Whisper models
- A free ASR software called “NoScribe” was/is available

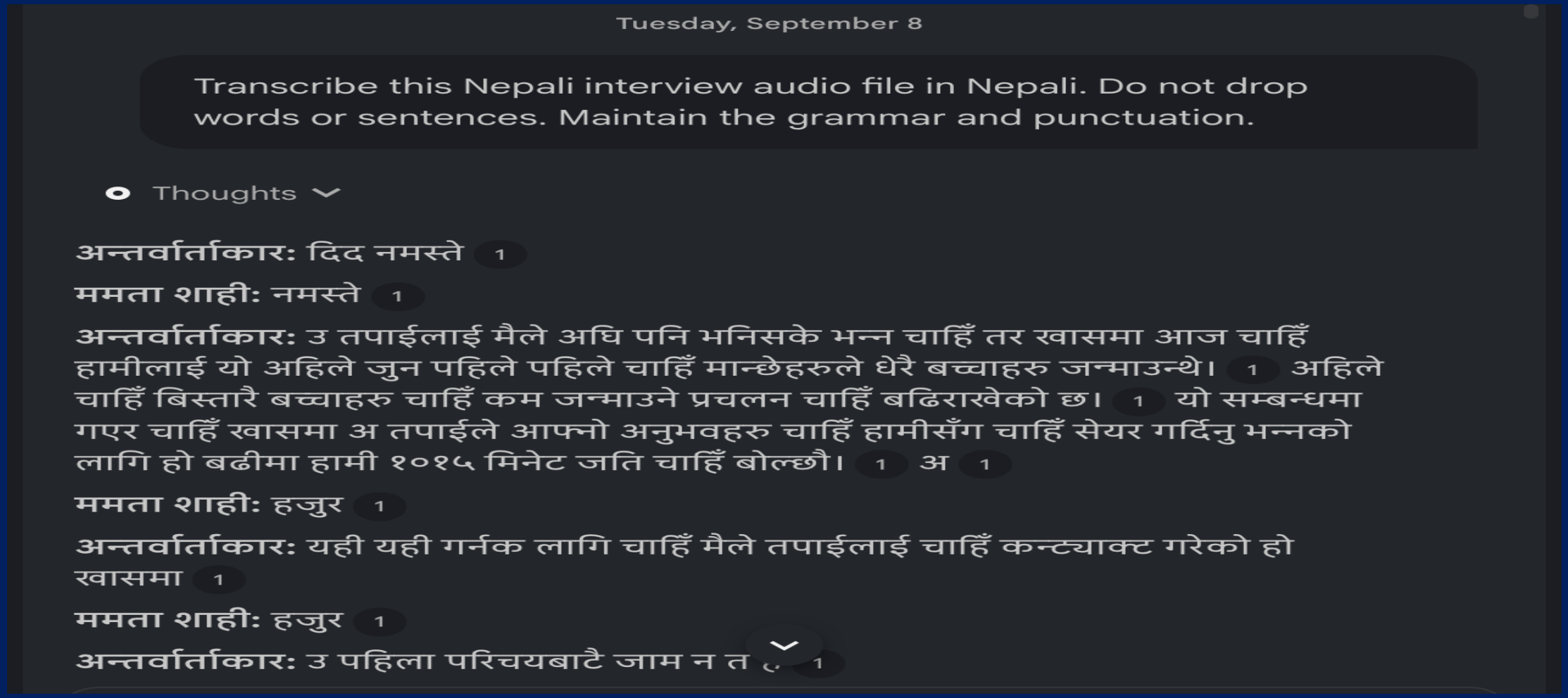
Present scenario 2

- Gemini Notebook (Notebook LM) can transcribe the Nepali Audio file (clear recordings) on the fly with around 90% accuracy
- The researcher can easily and timely correct the transcript file
- He/she can also ask the notebook to tranlate the file for further analysis i.e. quantitative or qualitative content analysis.
- The quantiative content analysis is also called text analytics.

Example Notebooks:



A Nepali audio file and its transcription



Pre-processing of Nepal audio file

- The transcribed file must be read throughly and changed must be done by the researcher
- Once the researcher confirms all the data then only it must undergo to the another process i.e. translation
- However, you/we can ask the Notebook to do the translation without pre-processing and the clean the translated file instead if and only if the time is an issue

Direct translation of the Nepali transcript

Good! Now translate this Nepali interview transcript to English. Do not drop words and sentences. Maintain grammar and punctuations.

Thoughts

Interviewer: Sister, Namaste 1 .

Mamata Shahi: Namaste 1 .

Interviewer: Um, I already told you earlier, but actually, today we... you know how in the past, people used to give birth to many children 1 ? Nowadays, the practice of gradually giving birth to fewer children is increasing 1 . Regarding this matter, actually, we want you to share your experiences with us 1 . We will speak for about 10 to 15 minutes at most 1 . Um...

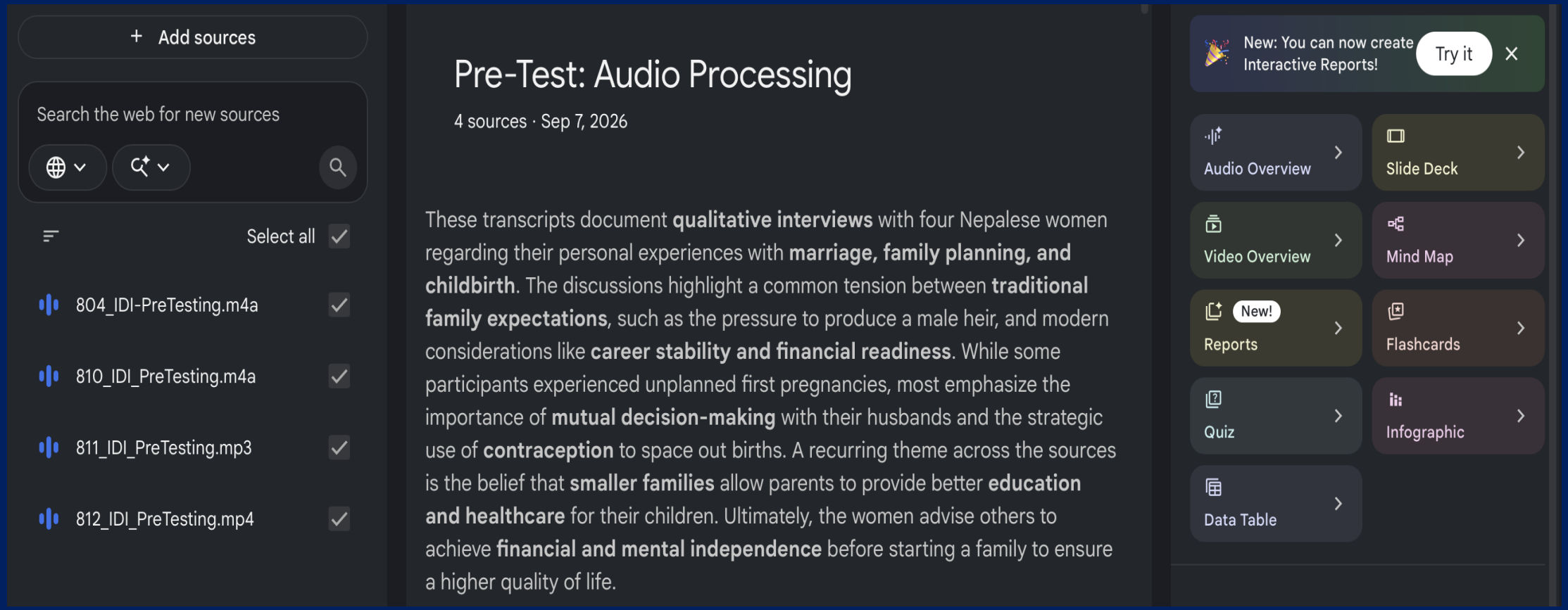
Mamata Shahi: Yes 1 .

Interviewer: This is exactly why I contacted you, actually 1 .

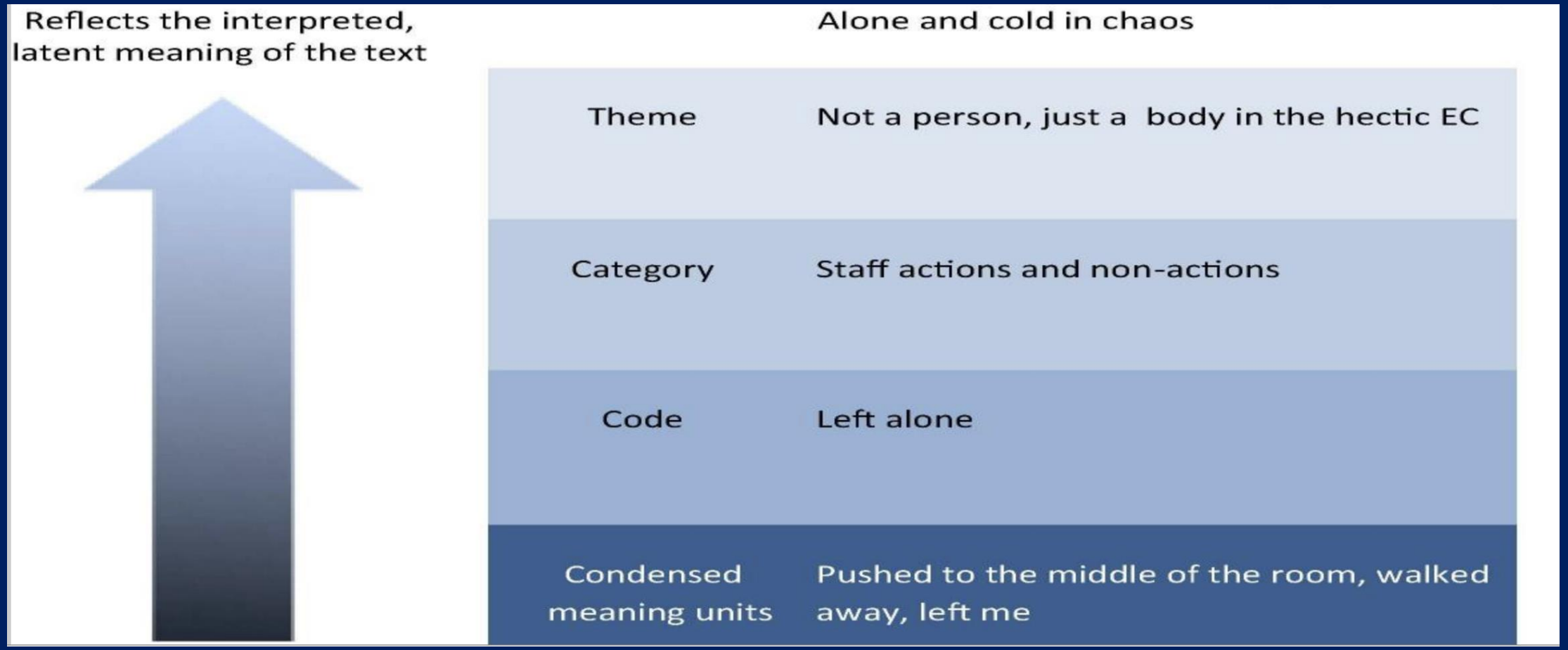
Mamata Shahi: Yes 1 .

Interviewer: Um, let's start with an introduction first, shall we 1 ?

This is the short summary after four Nepali audio files are processed: What next?



Option 1: Qualitative Content Analysis - From Condensed Meaning units to Theme



https://pmc.ncbi.nlm.nih.gov/articles/PMC6234169/

Option 2: Text to Corpus to Numbers to Graphs to find meanings

- Quantiative content analysis
- Here we convert the texts into “meaningful” numbers
- Do visualization with it
- Do sentiment analysis with it
- The translated files must be imported as a “text corpus” in R or Python
- Use “Text Mining” package such as “tm” available for R to pre-process the corpus
- Create TDM or DTM as needed

Text Mining with “Gemini Notebook”: Python

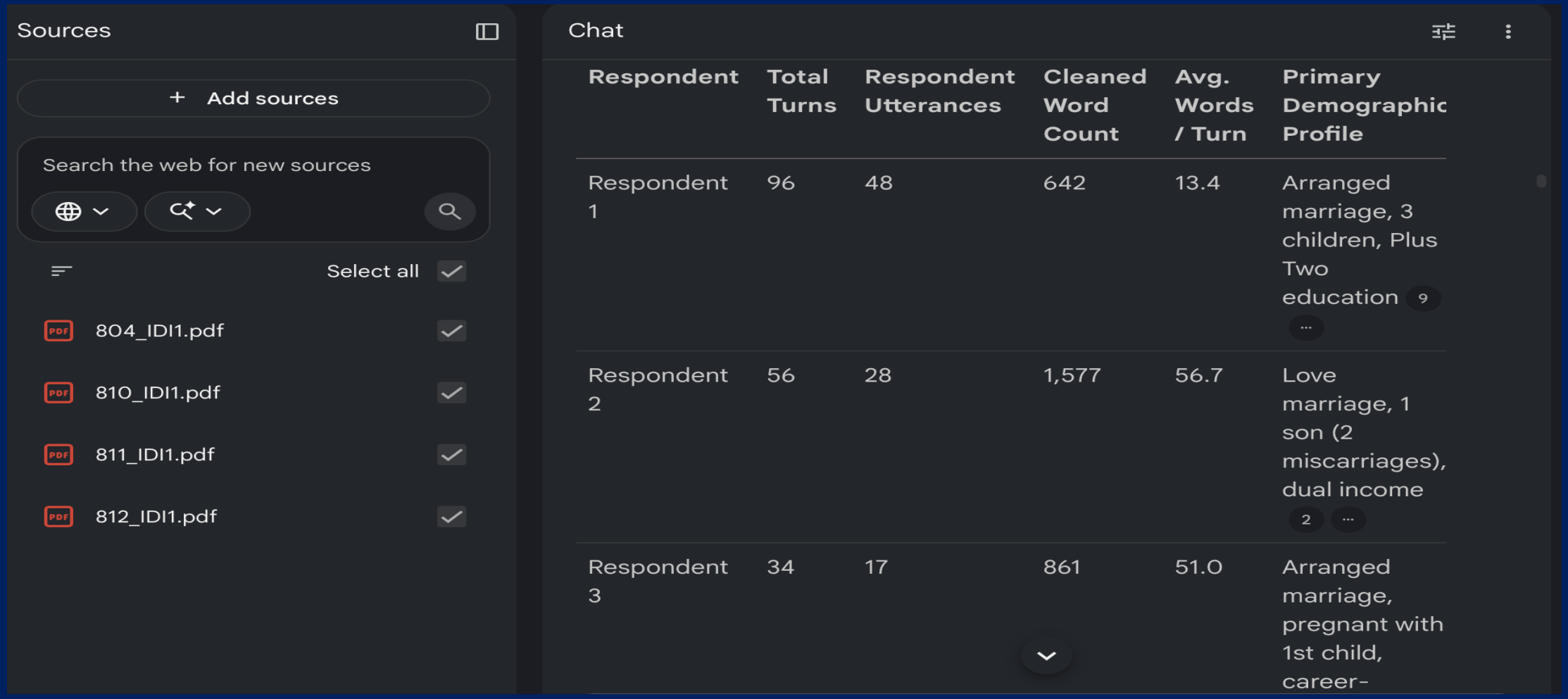
- Corpus summary Statistics:

- Total Interviews: 4 IDIs
- Total Conversational Turns: 269 turns
- Total Word Count: 8051 words
- Respondent Dialogue Words: 4883 (60.7%)
- Interviewer Dialogue Words: 3168 (39.3%)

- Unique vocabulary size: 884 disticnt tokens

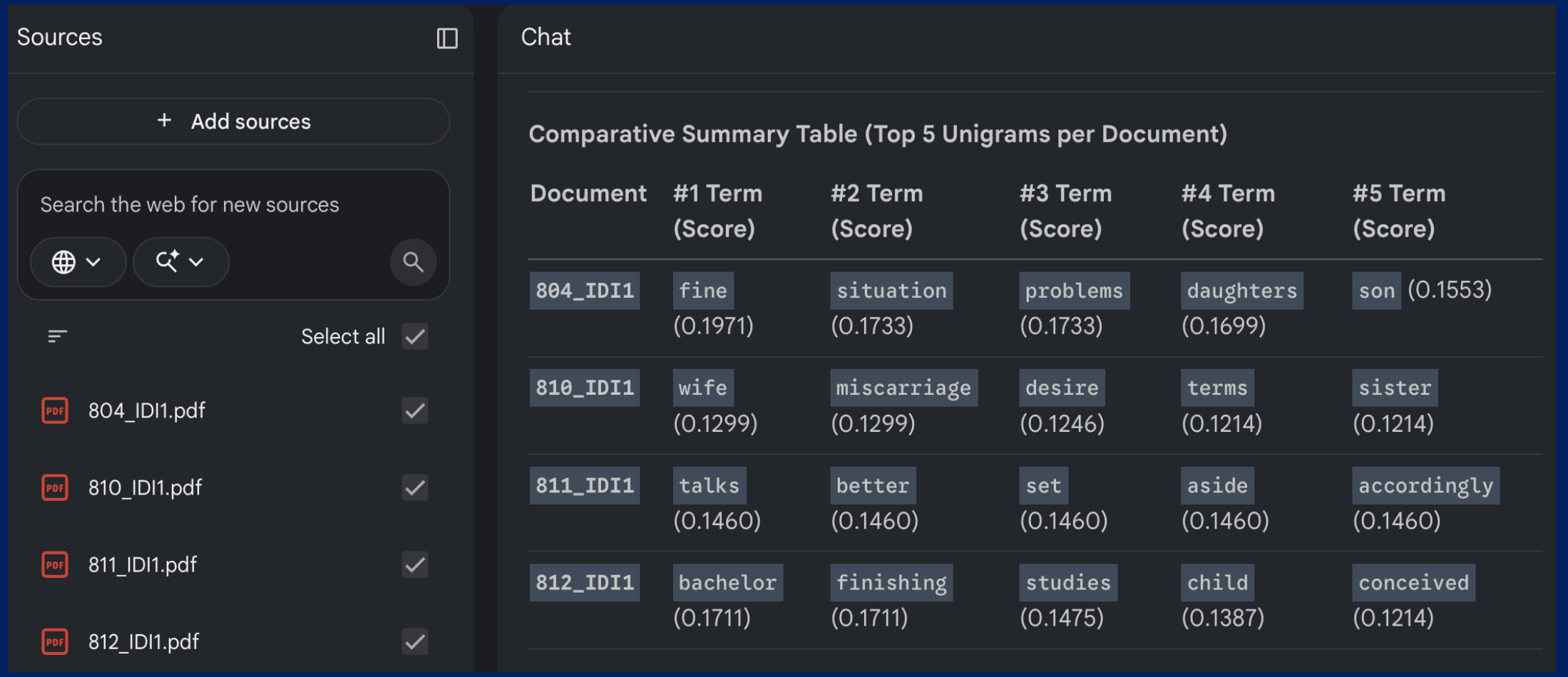
- Respondent Dialogue Key Terms:
- Child/marriage: 106 occurrences
- Time: 29 occurrences
- Married/marriage: 49 occurrences
- Husband/father/law: 63 occurences
- Plan/planning: 34 occurences
- Feel/desire/pressure: 31 occurences

Respondent wise summary statistics:



Respondent	Total Turns	Respondent Utterances	Cleaned Word Count	Avg. Words / Turn	Primary Demographic Profile
Respondent 1	96	48	642	13.4	Arranged marriage, 3 children, Plus Two education 9 ...
Respondent 2	56	28	1,577	56.7	Love marriage, 1 son (2 miscarriages), dual income 2 ...
Respondent 3	34	17	861	51.0	Arranged marriage, pregnant with 1st child, career-

Text Analytics: TF-IDF keyword extraction



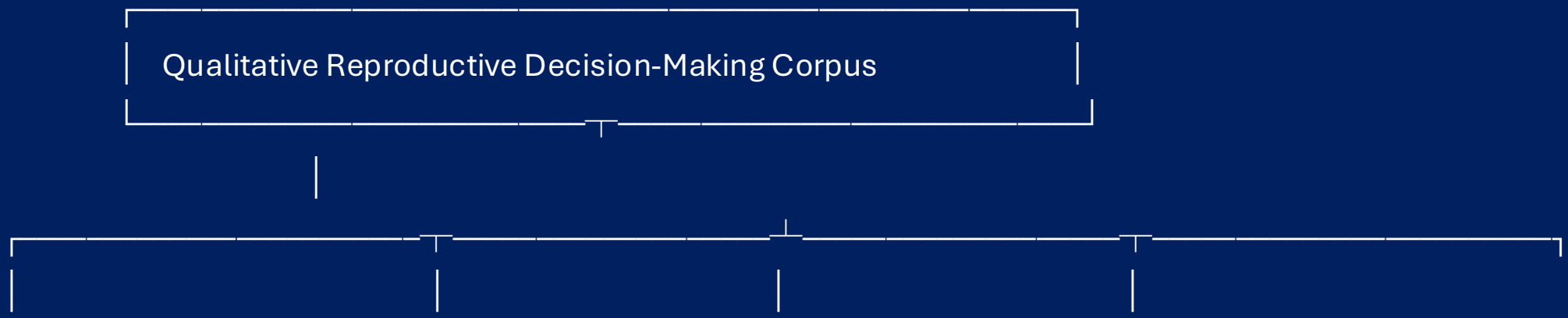
Text Analytics: TF-IDF keyword extraction

- A Term Frequency (TF)-Inverse Document Frequency (IDF) was used to find the Corpus-Wide Top **Distintive N-Grams**

• Father law = 0.0501	Decision making hierarcy and pressure for male heir
• Having child = 0.0493	Fertility timing and intention
• Mother law = 0.0416	Intra-household pressure regarding grandchildren
• Getting married = 0.0383	Marital age, Arrange vs Love marriage

Latent Dirichlet Allocation (LDA) topic model

- **LDA Latent Topic Model Overview**



- [Topic 1: 100%] Son Preference & In-Law Pressure (Respondent 1)
- [Topic 2: 99.9%] Dual-Career Jobs & Miscarriage Impact (Respondent 2)
- [Topic 3: 100%] Spousal Timing & Montessori Costs (Respondent 3)
- [Topic 4: 100%] Academic Milestones & Natural Family Planning (Respondent 4)

Sentiment analysis of the four transcripts

- **Overall Sentiment Polarity Distribution**



[Positive / Content]
- Spousal harmony
- Post-son relief
- Self-reliance

[Pragmatic / Cautious]
- Birth spacing & timing
- Economic cost evaluation
- Single-child preference

[Anxious / Stressed]
- In-law son pressure
- Societal judgment
- Educational expenses

[Grief / Frustration]
- Career lag due to kids
- Miscarriages & infection
- Environmental infertility

Read “corpus” and create term-document matrix in R/Python:

```
> print(tdm)
```

<<TermDocumentMatrix (**terms: 833**, documents: 4)>>
Non-/sparse entries : 1411/1921
Sparsity : 58%
Maximal term length : 22
Weighting : term frequency (tf)

Terms with frequencies > 10 occurences

```
> freq <- findFreqTerms(tdm, lowfreq = 10)
> freq
 [1] "able"       "affect"     "affected"   "age"        "already"    "another"
 [7] "anything"   "birth"      "bit"        "born"       "can"        "career"
[13] "case"       "child"      "children"   "currently"  "daughter"   "decision"
[19] "desire"     "dif"        "early"      "education"  "even"       "experience"
[25] "factors"    "family"     "fatherlaw"  "feel"       "fewer"      "gap"
[31] "gave"       "get"        "getting"    "give"       "giving"     "good"
[37] "got"        "happen"     "health"     "home"       "husband"    "indeed"
[43] "job"        "just"       "know"       "late"       "looking"    "love"
[49] "made"       "make"       "makes"      "many"       "marriage"   "married"
[55] "matter"     "mean"       "might"      "much"       "must"       "nancial"
[61] "nothing"    "now"        "number"     "okay"       "one"        "parentslaw"
```

Word Cloud from Term-Document Matrix:



Text Analytics done with Python/R:

- It is scalable
- It can be “automated”
- This will be handy for a getting a “quick” meaning from the valid and reliable text data
- Qualitative content analysis is needed to get a “rich” and “depth” analysis of the transcribed/translated files

Questions/queries?

Thank you!

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